

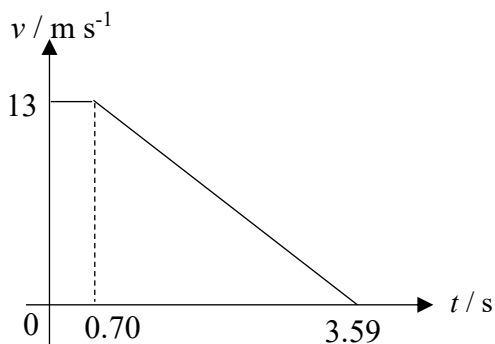
Q1(a)

Upon stepping on the brakes, $v = u + a t$

$$0 = 13 + (-4.5) t$$

$$t = \mathbf{2.89 \text{ s}}$$

1(b)



1(c)

Total displacement travelled from the time the motorist sights the red light till the time the car comes to a stop = area under the velocity-time graph

$$= 13 (0.70) + (1/2)(13)(2.89)$$

$$= 27.9 \text{ m}$$

$$\text{Distance from stop line} = 27.9 - 25 = \mathbf{2.9 \text{ m}}$$

1(d)

Find time for twig to fall vertically through 50 m:

$$s = u t + \frac{1}{2} a t^2$$

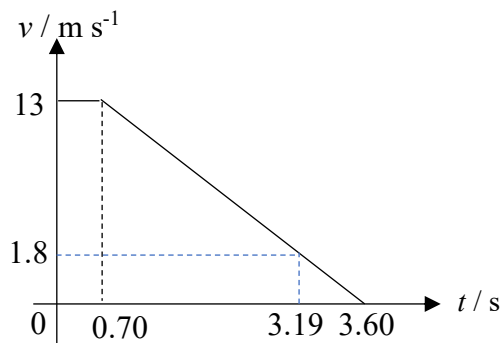
$$50 = 0 + \frac{1}{2} (9.81) t^2$$

$$t = 3.19 \text{ s}$$

$$\text{Horizontal distance travelled by twig in } 3.19 \text{ s} = 6.0 (3.19) = 19.1 \text{ m}$$

In 3.19 s, the velocity of the car is

$$v = 13 - 4.5(3.19 - 0.70) = 1.8 \text{ m s}^{-1}$$



Distance covered by the car is

$$13(0.70) + \frac{1}{2}(13 + 1.8)(3.19 - 0.70) \\ = 27.5 \text{ m}$$

Since the twig has travelled a shorter distance compared to the car, it will not hit the car.

Q2(a)

Power is work done per unit time or rate of doing work